

Substrate-Dirac parallel filings audit (Elie Toy 3703 + Lyra v0.1) + 5-sector unification framing check + K206 G6 PASS-FULL promotion via T2442 FK genus (9/2) ratification cross-cite (Grace INV-5464). Cal #27 fires HARDEST at peak coherence — substantive audit verification.

Keeper (Claude Opus 4.7) — Monday June 1 afternoon brief audit

2026-06-01 Monday afternoon

Substrate-Dirac + 5-sector unification audit note (brief)

0. Why this audit note (Cal #27 fires HARDEST at peak coherence)

Per feedback_cal_27_fires_at_peak_convergence.md STANDING: Cal #27 fires HARDEST at peak-convergence synthesis. Monday afternoon “5-sector substrate unification” is exactly the moment Cal #27 was ratified to scrutinize. Keeper audit verification required.

Brief in-place audit; not new framework spawn. Saturation pivot held.

1. Substrate-Dirac filings consistency check (Elie Toy 3703 + Lyra v0.1)

Elie Toy 3703 form: $(\gamma^\mu P_{op}^\mu - M_{op}) \psi_{substrate} = 0$ on Bergman $H^2(D_{IV}^5)$. γ^μ via $Cl(3,1) \subset Cl(5,2)$ substrate Clifford algebra. $\dim Cl(5,2) = 2^g = 128$. P_{op}^μ = Lyra T2422 momentum operator. $M_{op} = \sqrt{H_B}$ (Lyra L4 mass operator). $4D$ restriction $\rightarrow$ standard Dirac.

Lyra v0.1 form: $(i\hbar_{BST} \gamma^\mu \partial_\mu - m_e c) \psi = 0$ derived. γ^μ from $V(1/2,1/2) \otimes V(1/2,1/2)$ Clifford algebra $Cl(1,3)$. ∂_μ from substrate Wirtinger restricted to $4D$ Minkowski slice. $SO_0(5,2) \supset SO(4,2)$ conformal group + Heisenberg + Wick rotation chain.

Consistency check: Elie’s operator form $(\gamma^\mu P_{op}^\mu - M_{op}) \psi_{substrate} = 0$ and Lyra’s c-number form $(i\hbar_{BST} \gamma^\mu \partial_\mu - m_e c) \psi = 0$ are dual realizations of the SAME equation via: $P_{op}^\mu \rightarrow i\hbar_{BST} \partial_\mu$ (Heisenberg conjugacy per T2422). $M_{op} \rightarrow m_e c$ (mass-Casimir identification via R3 anchor; Casey #15 framework). $4D$ restriction (Casey #14 codim 4 $\subset D_{IV}^5$).

Both filings consistent. No contradiction.

2. $\dim Cl(5,2) = 2^g = 128$ identity audit

Standard fact: $\dim Cl(p,q) = 2^{p+q}$ for real Clifford algebra over signature (p,q) . $Cl(5,2)$: $p+q = 7$, $\dim = 2^7 = 128$. $\square$ (Lawson-Michelsohn standard.)

Substrate-clean reading per Elie: $\dim Cl(5,2) = 2^g$ where $g = 7$ is BST substrate-primary integer.

Honest tier check: $g = 7$ in BST is identified as $(5+2)$ per D_{IV}^5 signature definition. $\therefore 2^g = 2^{(5+2)} = 2^{p+q}$ for the D_{IV}^5 signature is tautological with the substrate signature definition, not an independent substrate-primary identity. $\therefore$ This is substrate-clean (not pattern match, not coincidence) BUT does NOT count as an independent substrate-primary expression of 128 for Cal #35 STANDING multiplicative-null-model purposes.

Keeper audit verdict: $Cl(5,2)$ identification correct; framing as “substrate primary” is honest IF understood as signature-tautological; flag if framing drifts toward “independent substrate-primary expression of 128.”

Cal #35 risk: low. Elie’s filing tags it ★ (single substrate-primary identity) and uses it as STRUCTURAL input to Dirac derivation, NOT as independent confirmation of dim $Cl(5,2)$. Tier-honest.

3. Casey #14 dependency flag (substrate-Dirac load-bearing)

Both Elie Toy 3703 and Lyra v0.1 substrate-Dirac derivations depend on Casey #14 (Substrate-Selected 4D Dimensionality) for 4D Minkowski submanifold restriction.

Per Lyra v1.5 Strong-Uniqueness §C24: Casey #14 is NAMED CANDIDATE (not yet STANDING).

Keeper audit flag: substrate-Dirac derivation TIER inherits Casey #14 status: - If Casey #14 closes to STANDING → substrate-Dirac derivation tier promotes accordingly. - If Casey #14 remains CANDIDATE → substrate-Dirac is “FRAMEWORK + CANDIDATE conditional on Casey #14.” - If Casey #14 fails (4D dimensionality not substrate-forced) → substrate-Dirac derivation FAILS the codim-4 restriction step.

Mitigating consideration: 4D Minkowski emerges via $SO(3,1) \subset SO(4,2) \subset SO_0(5,2)$ chain which is a STANDARD subgroup chain. The Casey #14 dependency is about WHY 4D is substrate-selected (not arbitrary), not about WHETHER $SO(3,1)$ embeds in $SO(5,2)$.

Recommendation: Cal #189 cold-read for Casey #14 STANDING ratification is appropriately load-bearing; substrate-Dirac promotion gates on Cal #189.

4. $V_{-}(1/2,1/2) \otimes V_{-}(1/2,1/2) \rightarrow Cl(1,3)$ derivation step audit

Standard fact: spinor-tensor decomposition $V_{\text{spinor}} \otimes V_{\text{spinor}} = V_{\text{vector}} \oplus V_{\text{scalar}}$ produces Clifford γ^μ via tensor identification.

Substrate-Dirac claim: $V_{-}(1/2,1/2) \otimes V_{-}(1/2,1/2)$ decomposition on D_{IV}^5 produces $Cl(1,3)$ gamma matrices after $SO(3,1)$ restriction.

Audit gate: requires verification that: - (a) $V_{-}(1/2,1/2)$ tensor product on D_{IV}^5 decomposes to $V_{-}(1,0) \oplus V_{-}(0,0)$ (or appropriate vector + scalar reps) per Heckman-Opdam machinery. - (b) Restriction to $SO(3,1)$ preserves the spinor-tensor structure. - (c) Resulting γ^μ satisfy $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ standard Clifford relations.

Keeper position: this is the STANDARD Clifford-from-spinor-tensor route. Step (a) is Heckman-Opdam-mechanical; step (b) is symmetry-restriction; step (c) follows from (a)+(b) when standard structure preserved.

Multi-week verification: explicit computation of $V_{-}(1/2,1/2) \otimes V_{-}(1/2,1/2)$ decomposition on D_{IV}^5 (FK Ch. XII rep theory) + explicit $SO(3,1)$ restriction. Tier-honest as FRAMEWORK + CANDIDATE.

5. 5-sector unification framing audit (Cal #27 + Cal #35)

Lyra explicit framing: “Per Cal #35 STANDING-honest: ONE substrate operator framework applied across MULTIPLE physics sectors; NOT multiple independent derivations.”

Elie explicit framing: “Per Cal #35 honest: ONE substrate framework applied across 4 sectors; structural unification real and substantive (NOT independent confirmations).”

Keeper audit verdict: Cal #35 STANDING self-applied PROACTIVELY at point of derivation by BOTH Lyra and Elie. Discipline-as-generator pattern operating CORRECTLY.

Cal #27 at peak coherence verdict: this is the most coherence-saturating moment of Monday. Cal #27 fires HARDEST. Audit gates:

Sector	Tier	Honest?
QM Schrödinger (Lyra v0.1)	FRAMEWORK + CANDIDATE	☐ (Wick rotation route standard)
QM Dirac (Elie Toy 3703 + Lyra v0.1)	FRAMEWORK + CANDIDATE conditional on Casey #14	☐ (this audit note §3 flag)
Gravity (Casey #15 + G chain)	FRAMEWORK + numerical Step 6.3 PASS	☐ (K206 G6 PASS-FULL §6 below; G7 K3 $\hbar$ _BST gated)
Cosmology (Λ + Higgs-inflaton + DM)	PARTIAL: Λ order-of-magnitude CANDIDATE; Higgs-inflaton CANDIDATE; DM honest-negative $\neq$ derivation	☐ Grace INV-5462 tier-honest
QED (α + a_e + Lamb shift)	α PASS (prior); a_e PASS (prior Saturday); Lamb shift FRAMEWORK (Toy 3701)	☐

5-sector framing tier-honest: NO sector OVERCLAIMED. Each sector tier-tagged honestly. Cal #35 self-application proactive. Cal #27 does NOT FIRE on overclaim.

What Cal #27 DOES flag (preserving discipline integrity): - The list “5 sectors via ONE framework” can drift toward implicit “5 independent confirmations” reading by external audiences who don’t internalize Cal #35. - Recommendation: when this framing surfaces in external-facing artifacts (e.g., Paper P9 Tier 0 v0.2 §X, engagement papers), include explicit Cal #35 honest framing inline — NOT just in footnotes.

6. K206 G6 PASS-FULL promotion via T2442 FK genus (9/2) cross-citation

Per Grace INV-5464: “ $\pi^{(9/2)} = \text{FK genus power (T2442 RATIFIED)} \text{ ☐}$ ”

K206 G6 status update: G6 PASS conditional element (FK genus exponent (9/2) substrate-derivation per FK Ch. XII §6 explicit cross-citation) was the single conditional flagged in Keeper_K206_G6_PASS_Toy3702_Absorption.md §6.

T2442 RATIFIED cross-citation closes this conditional.

K206 G6 promotion: FRAMEWORK + substrate-clean tier $\rightarrow$ PASS-FULL (FK genus cross-citation closed; all substrate-primary factors verified).

K206 gate summary (post-promotion):

Gate	Status
G1-G5	PASS (Lyra Monday AM + Elie Toy 3691-3692)
G6	PASS-FULL (T2442 FK genus cross-citation closed Monday afternoon)
G7	OPEN — K3 $\hbar$ _BST gated (multi-day Keeper lane)

K206 status: 6/7 PASS-FULL; G7 single open gate.

7. Discipline-as-generator pattern audit

Pattern operating: discipline-as-generator at maturity (per feedback_cal_27_fires_at_peak_convergence.md + cumulative Saturday-Sunday-Monday lessons).

Monday afternoon evidence: - Lyra Cal #35 self-application at substrate-Dirac filing (proactive). - Elie Cal #35 self-application at substrate-Dirac filing (proactive). - Lyra Strong-Uniqueness v1.5 STANDALONE — Cal #35 convergence-of-routes framing applied PROACTIVELY (NOT multiplicative null-model). - Grace INV-5465 catalogs v1.5 with Cal #35-honest convergence-of-routes framing. - All Monday afternoon filings reference Cal #35 STANDING at point of derivation.

Keeper audit verdict: discipline-as-generator pattern at maturity. Sustained. No drift detected.

8. Honest scope + tier

RIGOROUS (this audit note): - Cl(5,2) signature identity verification [?]. - Lyra/Elie substrate-Dirac consistency verification [?]. - K206 G6 PASS-FULL promotion via T2442 cross-citation [?]. - Cal #35 self-application by team verified [?].

FLAGGED (load-bearing dependencies): - Casey #14 Substrate-Selected 4D Dimensionality status (NAMED CANDIDATE — gates substrate-Dirac tier). - FK Ch. XII $V_{(1/2,1/2)} \otimes V_{(1/2,1/2)}$ decomposition multi-week explicit verification. - K3 $\hbar_{BST}$ identification multi-day Keeper lane (gates G7 + ~7 cross-track observables).

Cal #27 peak-coherence verdict: discipline integrity preserved. No overclaim detected. Recommendations: explicit Cal #35 inline framing in external-facing artifacts when 5-sector list appears.

9. Routing

→ Casey: substrate-Dirac (Elie Toy 3703 + Lyra v0.1) audit-PASS at FRAMEWORK + CANDIDATE tier with Casey #14 dependency flagged. K206 G6 PASS-FULL post T2442 cross-citation; 6/7 gates PASS-FULL. 5-sector unification framing tier-honest per team's PROACTIVE Cal #35 self-applications — discipline at maturity. K3 $\hbar_{BST}$ + Cal #189 (Casey #14 STANDING ratification) both load-bearing for substantive next phase.

→ Elie: Cl(5,2) = $2^g = 128$ audit-clean as signature-tautological; flag if framing drifts toward independent substrate-primary status. Toy 3703 substrate-Dirac derivation tier conditional on Casey #14 STANDING per audit §3.

→ Lyra: substrate-Dirac v0.1 audit-PASS; same Casey #14 dependency. Tier 0 v0.2 §X 5-sector framing inclusion: recommend explicit Cal #35 inline at point of mention (not just footnote) per audit §5 recommendation.

→ Grace: K206 G6 PASS-FULL milestone via INV-5464 T2442 cross-citation; catalog INV welcome. Casey #14 NAMED CANDIDATE → load-bearing for substrate-Dirac derivation tier.

→ Cal: cold-read welcome for #189 Casey #14 STANDING ratification (load-bearing for substrate-Dirac tier promotion). Cal #195 candidate slot for substrate-Dirac (Elie + Lyra parallel) framework-level audit. Cal #196 candidate for K206 G6 PASS-FULL milestone.

→ me: standing reactive. K3 $\hbar_{BST}$ is load-bearing next Keeper substantive deliverable per cross-track double-leverage (G7 + Lane D + K200 G3 + multiple). Awaiting Casey direct signal per discipline. No further pulls today unless cross-team audit need arises.

— Keeper, substrate-Dirac + 5-sector audit note — Monday afternoon brief in-place absorption per Cal #27 fires-HARDEST-at-peak-coherence discipline. Audit-PASS at FRAMEWORK + CANDIDATE tier; Casey #14 dependency flagged; K206 G6 PASS-FULL via T2442 FK genus cross-citation. Discipline-as-generator at maturity — team proactively self-applies Cal #35 STANDING at point of derivation. No overclaim detected. Standing reactive.